Who Connects the Departments? Dependence and Resilience in the SNAP email-Eu-core Network

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***Abstract*—We study dependence and resilience in the SNAP email-Eu-core network of 1,005 employees across 42 departments. We measure department-pair dependence and compare random removal with targeting by external contacts or directed betweenness. Removing 100 employees by contacts retains 23.91% of cross-department edges, versus 81.54% for random removal. Betweenness targeting leaves lower survivor reachability than contact targeting (65.14% versus 67.51%). Many indirect paths remain after most direct cross-department edges are removed.**

***Index Terms*—complex networks, organizational communication, network resilience, node removal, betweenness centrality**

I. Introduction and Motivation

Departments communicate through individual employees. When many contacts involve the same person, that person's departure could remove most direct links between two departments. Other employees may still communicate through indirect paths, so counting lost contacts alone does not tell us how much connectivity remains.

We study this problem using the SNAP *email-Eu-core* network [1]. Our goal is to identify concentrated contacts between departments and measure how employee removal affects communication paths. We first measure dependence for each ordered department pair. We then rank employees by their number of contacts outside their department and by directed betweenness centrality. Finally, we compare targeted removal using these rankings with random removal at the same budgets. This lets us examine whether employees with many direct contacts are also the ones whose removal causes the greatest loss of directed reachability.

II. Previous Work

Yin et al. [2] use network motifs to find local communities and evaluate their method on email-Eu-core. Their analysis asks which groups can be recovered from network structure. We take the supplied department labels as fixed and ask how contacts between those groups depend on individual employees. This shifts the focus from finding communities to studying connections between them.

Brandes [3] gives an $O(NM)$ algorithm for exact betweenness centrality in unweighted graphs, where N and M are the numbers of nodes and edges. We use its NetworkX implementation [4]. Betweenness measures shortest-path brokerage across the whole graph; our contact score counts distinct neighbors outside an employee's department. Comparing both rankings and their removal outcomes tests whether these two roles have different effects on connectivity.

Albert et al. [5] show that scale-free networks can tolerate random failures while being vulnerable to targeted removal of high-degree nodes. We use the same random-versus-targeted comparison without assuming that email-Eu-core is scale free. Our evaluation separates the loss of direct cross-department edges from the loss of directed paths between employees in different departments. Measuring both matters because indirect paths may survive after many direct edges disappear.

Ghosh and Boyd [6] study how to add a limited number of edges to improve algebraic connectivity in undirected graphs. Their spectral objective does not directly apply to our directed reachability measure. We use their budgeted edge-selection setup as a reference for planned experiments, not as a method in the current analysis. We will compare backup contacts with random additions under equal budgets using employee-level directed paths.

III. Approach

A. Dataset and Network Construction

The SNAP *email-Eu-core* dataset contains internal email relationships at a European research institution [1]. Each node is an employee, and a directed edge (u, v) means that u sent at least one email to v. The dataset has $1,005$ employees, 42 departments, and $25,571$ directed edges. Each employee has one department label. Department sizes range from 1 to 109 employees, with a median of 14.5.

We build an unweighted directed graph $G = (V, E)$ and check that all edge endpoints have department labels. Removing 642 self-loops leaves $24,929$ edges: $16,284$ cross-department edges (65.3%) and $8,645$ internal edges (34.7%). We retain all labeled employees, including the 19 nodes that become isolated after removing self-loops. The largest weakly

and strongly connected components contain 986 and 803 nodes, respectively.

B. Research Questions and Mathematical Formulation

We address three research questions:

RQ1: Concentration of Department-Pair Contacts. For any two distinct departments A and B, let $E_{AB} \subseteq E$ denote the set of edges from employees in A to employees in B. For each employee $v \in A \cup B$, let $c_{AB}(v)$ denote the number of edges in E_{AB} incident to v. For pairs with $|E_{AB}| \geq 1$, we define the directed department-pair dependence D_{AB} as:

$$D_{AB} = \frac{\max_{v \in A \cup B} c_{AB}(v)}{|E_{AB}|} \in (0, 1]. \tag{1}$$

A value of $D_{AB} = 1$ means that every observed edge from A to B involves one employee. For example, four edges from one sender in A to four receivers in B give $D_{AB} = 1$. Four edges with four different senders and four different receivers give $D_{AB} = 1/4$. These are counts of distinct relationships, not email messages. Since a single-edge pair always has $D_{AB} = 1$, we compare minimum edge-support thresholds $k \in \{1, 5, 10\}$. High direct dependence alone does not establish that removing the employee would eliminate indirect paths.

RQ2: Local Contacts vs. Global Path Brokerage. Let $d(v)$ denote employee v's department. The cross-department contact score counts distinct neighbors outside that department, taking the union of incoming and outgoing contacts:

$$\begin{aligned} \mathcal{N}_{\text{cross}}(v) = \{u \in V : d(u) \neq d(v), \\ (u, v) \in E \ \vee \ (v, u) \in E\}, \\ k_{\text{cross}}(v) = |\mathcal{N}_{\text{cross}}(v)|. \end{aligned} \tag{2}$$

We also compute normalized directed betweenness [3]:

$$C_B(v) = \frac{1}{(N-1)(N-2)} \sum_{\substack{s,t \in V \setminus \{v\} \\ s \neq t}} \frac{\sigma_{st}(v)}{\sigma_{st}}, \tag{3}$$

where $N = |V|$, σ_{st} is the number of directed shortest paths from s to t, and $\sigma_{st}(v)$ counts those passing through v. Unreachable pairs contribute zero, and endpoints are excluded. We compare the rankings to see whether employees with many external contacts also lie on many shortest paths.

RQ3: Network Resilience under Removal. How rapidly does cross-department communication degrade when key personnel are removed? We benchmark three removal strategies: (1) Random removal (30 Monte Carlo permutations), (2) Targeted removal by descending cross-department contact count, and (3) Targeted removal by descending directed betweenness.

C. Assumptions

We treat the network as static and unweighted. The data record observed relationships without message counts or timestamps, so we cannot measure communication frequency or the timing of information flow. Edges retain their direction: $u \to v$ does not imply $v \to u$. A directed path indicates possible structural reachability; it does not show that information actually traveled along that path.

Removal deletes an employee and all incident edges. Rankings stay fixed at their values in the original graph. We do not model employees forming new contacts after a departure. All reachability calculations use paths in the employee graph, which avoids introducing paths through department aggregation.

IV. Experimental Setup and Results

A. Experimental Setup

We evaluate removal budgets of $B \in \{0, 10, 20, 50, 100\}$ employees, obtained by rounding down 0%, 1%, 2%, 5%, and 10% of the 1,005-node workforce. For each random seed from 0 to 29, we shuffle all employees and remove prefixes of the same order at increasing budgets. Targeted strategies use descending scores, breaking ties by ascending employee ID. Each experiment starts from a fresh copy of the original graph.

We report three metrics:

1) **Cross-department edge retention:** The ratio of surviving cross-department edges to the initial $16,284$ edges.
2) **Survivor reachability ratio:** The fraction of ordered cross-department employee pairs (u, v) with $u, v \in V_{\text{surviving}}$ and $d(u) \neq d(v)$ that possess a directed path.
3) **Reachability vs. original pairs:** The count of reachable surviving cross-department pairs divided by the initial $961,932$ cross-department pairs. This prevents denominator shrinkage from masking path destruction.

We count reachable pairs exactly using the directed acyclic graph (DAG) formed by condensing strongly connected components (SCCs). Nodes within an SCC are mutually reachable; paths between SCCs determine the remaining reachable pairs. Before removal, $755,692$ of $961,932$ ordered cross-department pairs are reachable (78.56%). The third metric uses all original eligible pairs as its denominator, so its baseline is also 78.56%, not 100%.

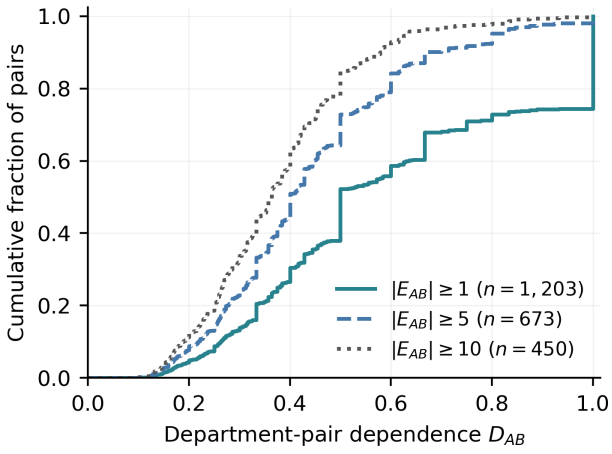


Fig. 1. Empirical cumulative distributions of department-pair dependence at three minimum edge-support thresholds. Each curve gives the fraction of retained ordered department pairs with dependence at most D_{AB}. Filtering sparse pairs reduces the share with complete dependence; see Table I.

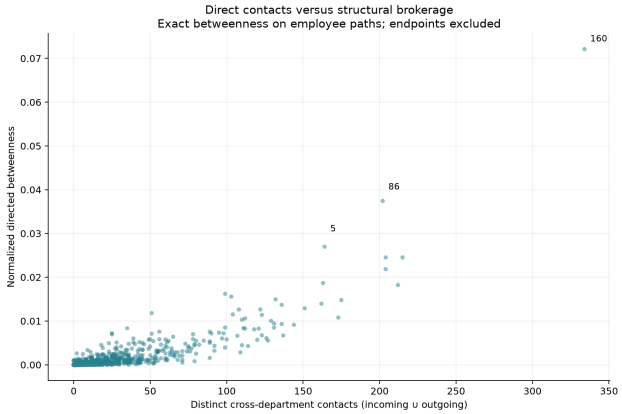


Fig. 2. Cross-department contacts vs. exact normalized directed betweenness for all 1,005 employees. The rankings overlap but also select different employees, as discussed in the text.

B. Preliminary Results

High pairwise dependence with substantial edge support. Of the $42 \times 41 = 1,722$ possible directed department pairs, $1,203$ have at least one edge. Their median dependence is 0.50, and 308 have $D_{AB} = 1$. Figure 1 shows how the dependence distribution changes when sparse pairs are excluded. The median falls from 0.50 to 0.40 and 0.364 at thresholds of five and ten edges, respectively (Table I). Complete dependence remains for 13 pairs with at least five edges. For Department $32 \to 14$, all 27 edges involve Employee 971. Removing this employee would remove every direct edge in that direction, although indirect paths may remain. For Department $24 \to 36$, Employee 163 is incident to 14 of 15 edges ($D_{AB} = 0.933$).

TABLE I
DEPARTMENT DEPENDENCE UNDER SUPPORT THRESHOLDS

Support (k)	Pairs Retained	Pairs with $D_{AB} = 1$	Median D_{AB}
$\lvert E_{AB}\rvert \geq 1$	1,203	308	0.500
$\lvert E_{AB}\rvert \geq 5$	673	13	0.400
$\lvert E_{AB}\rvert \geq 10$	450	1	0.364

Divergence between contacts and path brokerage. Employee 160 (Department 36) ranks first in both metrics, with 334 cross-department contacts and directed betweenness of 0.07212 (Fig. 2). The top-10 lists share eight employees. Employee 377 ranks 9th in betweenness and 42nd in contacts; Employee 129 ranks 10th and 39th, respectively. The rankings therefore select some different removal candidates.

Resilience under node removal. At $B = 100$, random removal retains an average of 81.54% of cross-department edges and leaves 77.22% of surviving cross-department pairs reachable (Table II). Contact-targeted removal retains 23.91% of edges, compared with 28.51% under betweenness-targeted removal. Betweenness targeting leaves lower survivor reachability: 65.14%, compared with 67.51% for contact targeting, a difference of 2.37 percentage points. The same ordering holds when reachable pairs are divided by the original pair count. Figure 3 shows the results across all removal budgets, with ± 1 sample standard deviation for random trials.

TABLE II
RESILIENCE METRICS AT 100 EMPLOYEES REMOVED (9.95% WORKFORCE)

Strategy	Edge Ret.	Surv. Reach.	Orig. Reach.
Random mean (30 trials)	81.54%	77.22%	62.61%
Contact-Targeted	23.91%	67.51%	54.52%
Betweenness-Targeted	28.51%	65.14%	52.74%

C. Discussion and Limitations

Contact targeting selects employees with many external neighbors, which helps explain the greater loss of cross-department edges. Betweenness targeting selects employees on many shortest paths and leaves fewer reachable pairs in these experiments. Even after contact targeting removes over 76% of cross-department edges, 67.51% of ordered cross-department pairs among surviving employees still have a directed path. This suggests that alternative paths preserve some connectivity after direct contacts disappear.

A remaining question is how dependence for a particular department pair relates to this global result. Our current experiments remove employees using global rankings; they do not establish the effect of removing the main contact for every highly dependent department pair. We will test selected pairs separately. The static graph also cannot capture changes in communication after employees leave. Our results describe structural connectivity in the observed network, rather than actual organizational performance.

V. CONCLUSION AND SHORT-TERM PLANS

A. Summary of Main Findings

Direct contacts between some departments are concentrated around one employee, even when the pair has many observed edges. Contact counts and directed betweenness produce overlapping rankings, but their removal effects differ. Contact targeting removes more direct cross-department edges, while betweenness targeting leaves lower directed reachability. We therefore use both measures when assessing resilience.

B. Plans for the Final Presentation

In early to mid November, we will study selected highly dependent department pairs. We will remove their main contact employees and measure both the direct edges lost and the remaining employee-level directed paths between the departments. This will help connect local dependence to the global removal results.

We will then define a rule for adding directed employee-level edges to provide backup connections. We will compare targeted additions with random additions under equal budgets of new cross-department edges, excluding self-loops and duplicate additions. Edges will be selected using pre-removal

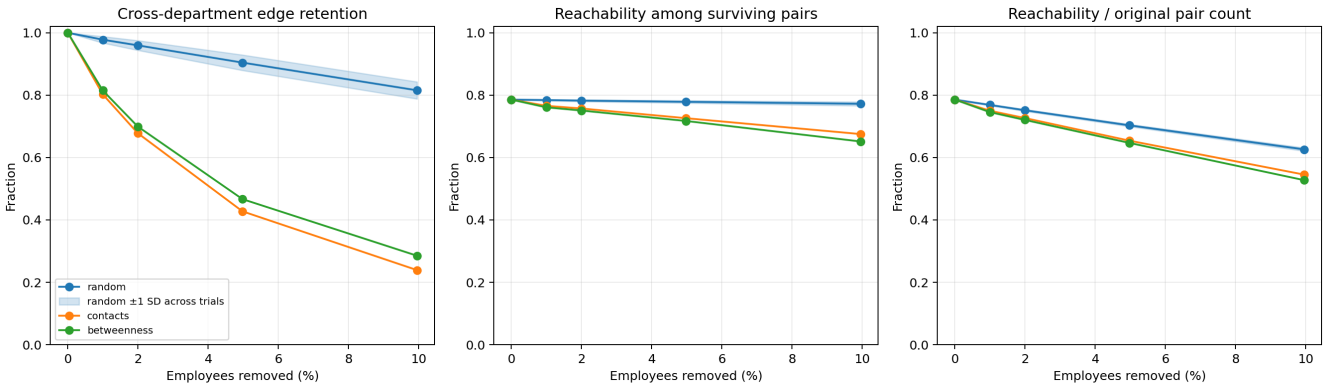


Fig. 3. Comparative resilience under random, contact-targeted, and betweenness-targeted employee removal across 0% to 10% workforce removal budgets. (Left) Cross-department edge retention; (Center) Directed reachability among surviving pairs; (Right) Reachable pairs normalized by the initial pair count $(961,932)$. Shaded bands indicate ± 1 sample standard deviation across 30 random trials.

information and added before removal. For each removal strategy and budget, all intervention variants will use the same employee removal list from the original graph, including identical random seeds; rankings will not be recomputed after adding edges. Edge retention will be measured relative to each augmented graph's pre-removal cross-department edge count. The reachability denominators will remain the surviving eligible pair count and the original $961,932$ eligible pairs, respectively. Ghosh and Boyd's edge-selection study [6] will inform this design. By November 30, we plan to complete these comparisons and prepare the final figures for the presentation in the first week of December. If time permits, we will also examine dynamic reranking and degree-preserving null models as separate sensitivity analyses.

C. Member Contributions

Jiening Yang handled data loading and cleaning, graph statistics, department-pair dependence, exact directed betweenness, employee rankings, and the corresponding plots. Jiening also prepared the analysis documentation and initial slides.

Yikang (Bruce) Cheng implemented the removal experiments and exact reachability calculation using SCC condensation, ran the random trials, compared resilience metrics, and prepared the removal plots and automated tests.

Both members will work on the department case studies, edge-addition comparisons, final report, and presentation.

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